

TANSHENG ZHU

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EDUCATION

Tsinghua University

Aug. 2025 – Present

Ph.D. Student in Computer Science, Institute for Interdisciplinary Information Sciences

Beijing, China

*Advisor: Prof. Jian Li***Shanghai Jiao Tong University**

Aug. 2021 – June 2025

B.S. (Honors) in Mathematics & Minor in Finance, Zhiyuan College

Shanghai, China

Advisor: Prof. Zhenli Xu

- Shanghai Outstanding Graduate (Top 1%); Best Bachelor Thesis of SJTU (Top 1%)
- Research Focus: *Bayesian Optimization, Monte Carlo Tree Search*

The Chinese University of Hong Kong

Jan. 2024 – Sept. 2024

Exchange Student | Junior Research Assistant, Department of Mathematics

Hong Kong, China

Advisor: Prof. Fenglei Fan

- Research Focus: *Deep Learning Theory, Dynamical Systems*

RESEARCH INTERESTS

My research interests center on the **mathematical foundations of deep learning and large language models**. I seek to understand the scaling behavior, optimization process, and implicit biases of neural networks from a mathematical perspective, with particular attention to the **interplay between model architecture and training dynamics**. I am also interested in statistical learning theory and its applications to modern AI systems.

My goal is to develop principled frameworks that not only explain the empirical success of deep learning but also guide the design of more **interpretable, reliable, and efficient** models.

PUBLICATIONS

AdaRoPE: Not All Attention Heads Should Rotate and Scale Equally*Shaowen Wang*, Yuke Zheng*, **Tansheng Zhu***, Shuang Chen*, Shaofan Liu, Suncong Zheng, Jian Li*

ICML 2026.

When Bias Pretends to Be Truth: How Spurious Correlations Undermine Hallucination Detection in LLMs*Shaowen Wang*, Yiqi Dong*, Ruinian Chang*, **Tansheng Zhu***, Yuebo Sun, Kaifeng Lyu, Jian Li*

ICLR 2026 Workshop on Principled Design for Trustworthy AI. arXiv:2511.07318

Bayesian Optimization by Kernel Regression and Density-based Exploration***Tansheng Zhu**, Hongyu Zhou, Ke Jin, Xusheng Xu, Qiufan Yuan, Lijie Ji*

Computational Optimization and Applications, 2026. arXiv:2502.06178

Bayesian-Optimized Progressive Widening via Kernel Regression for Monte Carlo Tree Search in Continuous Action Spaces*Ke Jin, **Tansheng Zhu**, Hongyu Zhou, Xusheng Xu, Qiufan Yuan, Lijie Ji*

23rd IFAC Symposium on Automatic Control in Aerospace (ACA 2025).

SELECTED AWARDS

Winning Prize in S.-T. Yau College Student Mathematics Contest, Appl. & Comp. Mathematics Track	May 2024
Finalist Winner in Mathematical Contest in Modeling (Top 3% of 18000+ teams)	May 2024
1st Prize in Shanghai Region of Chinese Mathematics Competitions	Nov. 2023
1st Prize in Huashu Cup National Mathematical Contest in Modeling (solo winner)	Aug. 2023
1st Prize in Chinese High School Mathematics League	Oct. 2020

SKILLS

Language: Chinese/Mandarin (native), English (fluent), Hokkien (native)**Programming:** Python (NumPy, SciPy, Sklearn, PyTorch), MATLAB, C/C++**Leadership:** Leadership Team Member, *Mathematical Modelling Association of SJTU*

2022 – 2024

Class President, *Zhiyuan College, Shanghai Jiao Tong University*

2021 – 2023